Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/17/2025
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for International Research Participation on ISS Kibo Module
Start Date:	05/01/2020	**End Date:**	04/30/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Phillips, Brandon	**Contact Phone:**	
Contact Email:	brandon.s.phillips@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2027 per F. Hernandez/BPS and NASA documentation (Ed., 4/21/26). NOTE: End date changed to 4/30/2026 per M. Sansoucie/MSFC (Ed., 5/2/25) NOTE: End date changed to 4/30/2025 per NSSC information (Ed., 10/22/22) NOTE: End date changed to 4/30/2022 per NSSC information (Ed., 5/14/21)		
Key Personnel Changes/Previous PI:	Per NASA, Grant #80NSSC26K0090 is a continuation of Grant 80NSSC20K1038, effective 5/1/2026 (Ed., 4/21/26).		
COI Name (Institution):	Lee, Jonghyun Ph.D. (Iowa State University) Watanabe, Masahito Ph.D. (JAXA PI: Gakushuin University, Japan)		
Grant/Contract No.:	80NSSC20K1038 ; 80NSSC26K0090		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: This is continuation of "Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops," grant NNX14AH85G, with the same Principal Investigator, Dr. Robert Hyers and co-Investigator, Prof. John Lee, now of Iowa State University. Funding is for Dr. Hyers's role as U.S. Principal Investigator (PI) on Professor Masahito Watanabe's (Department of Physics, Gakushuin University, Tokyo, Japan) proposal, “Interfacial Phenomena and Thermal Properties of High-Temperature Melts,” funded by the Japanese Aerospace Exploration Agency (JAXA). Proposed scope of work is summarized below: 1. Extend present numerical model supporting interfacial tension measurement experiments using the International Space Station-Electrostatic Levitation Furnace (ISS-ELF) and interpretation of the experimental data downloaded from the ISS to 3-D. The full three-dimensional model will provide a systematic understanding on the influence of experimental parameters on the measured interfacial tension. 2. Apply the model to different experimental parameters. These parameters include the initial amplitude of oscillation, the core-shell radii ratio, the phase difference of core and shell oscillation and sample rotation. 3. Design, direct, and analyze experiments to be conducted in the Marshall Space Flight Center (MSFC) Electrostatic Levitation Facility, as required. Prof. Hyers is the Principal Investigator. He will maintain overall responsibility for the project. Assisted by the postdoc in years 2-5, he will also contribute to the modeling efforts led by Prof. Lee at Iowa State.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the project are expected to be utilized for more efficient production of high-purity steels, which are the main metals used on Earth.
Task Progress:	• Ph.D. student, Ali Rabeh, was supported by the project and graduated. Now Rabeh is a postdoc at Iowa State. • Ph.D. student, Brayden Berg, is being supported by the project. • A two-phase model (single-phase drop + atmosphere) has been successfully developed with an error of ~3%. o Paper 1 (projection-method two-phase flows) is under review (Physics of Fluids). In this paper, we have successfully demonstrated the capability of a projection-based Cahn-Hilliard Navier-Stokes (CHNS) framework to model very high-density ratio (104 - 105 : 1) two-phase flows, specifically focusing on the dynamics of molten metal droplets in microgravity. Our approach leverages a thermodynamically consistent diffuse interface model, which has shown robust performance in handling the large density and viscosity ratios characteristic of these systems. • A three-phase model (core-shell two-phase drop + atmosphere). o Paper 2 (projection-method three-phase flows): the manuscript is being written. The two-phase model in Paper 2 has been extended to three-phase (core-shell) drop oscillation. Three validation cases were modeled: floating oil lens, bubble rise in stratified flow, and drop in shearing flow. The three-phase model is being extended to the core-shell model.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Articles in Peer-reviewed Journals	Lee J, Hyers RW. "Numerical estimation of convection in a molten zirconium droplet processed by electromagnetic levitation in microgravity." International Journal of Microgravity Science and Application. 2015 Jun;32(1):320105. https://doi.org/10.15011/jasma.32.1.320105 , Jun-2015